Fake Job Posting Detection

Machine Learning & NLP End-to-End Project Report

1. Executive Summary

Online recruitment fraud is a growing cyber threat targeting job seekers.
This project utilizes the EMSCAD dataset (17,880 listings) to build an automated machine learning system capable of identifying fraudulent job postings with high precision and recall.

2. Key Dataset Insights

- Total Postings: 17,880 | Legitimate: 17,014 (95.1%) | Fraudulent: 866 (4.9%)
- Class Imbalance: Extreme imbalance requires class-weight balancing and recall-focused metrics.
- High Missing Data: 'salary_range' (84% missing) was dropped. Metadata columns filled with 'Unknown'.

3. Model Performance Summary

Model	Accuracy	Precision (Fraud)	Recall (Fraud)	F1-Score	
Logistic Regression (Balanced)	96.25%	57.14%	90.17%	0.70	[BEST]
Multinomial Naive Bayes	95.83%	50.34%	84.97%	0.63	

4. Primary Signals of Fraud

- Missing Company Logo: Listings without a logo have significantly higher fraud rates.
- Absence of Screening Questions: Scammers avoid multi-step application screening.
- Text Vocabulary: High occurrence of terms like 'wire', 'data entry', 'earn money', 'urgently'.
- Shorter Descriptions: Median word count in fraudulent jobs is notably shorter than real positions.

Figure 1: Missing Values by Column (%)

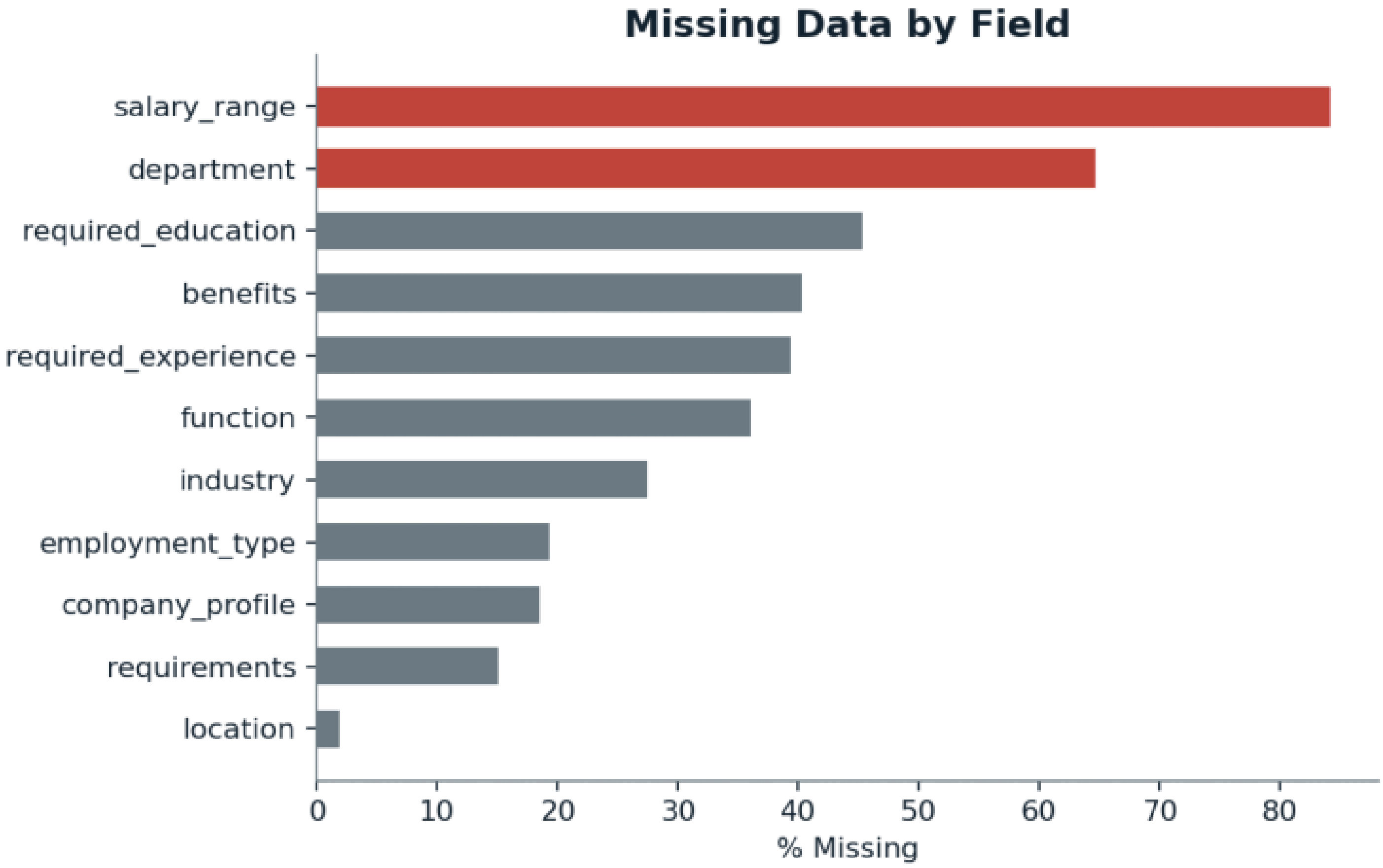

Figure 2: Impact of Text Cleaning & Preprocessing

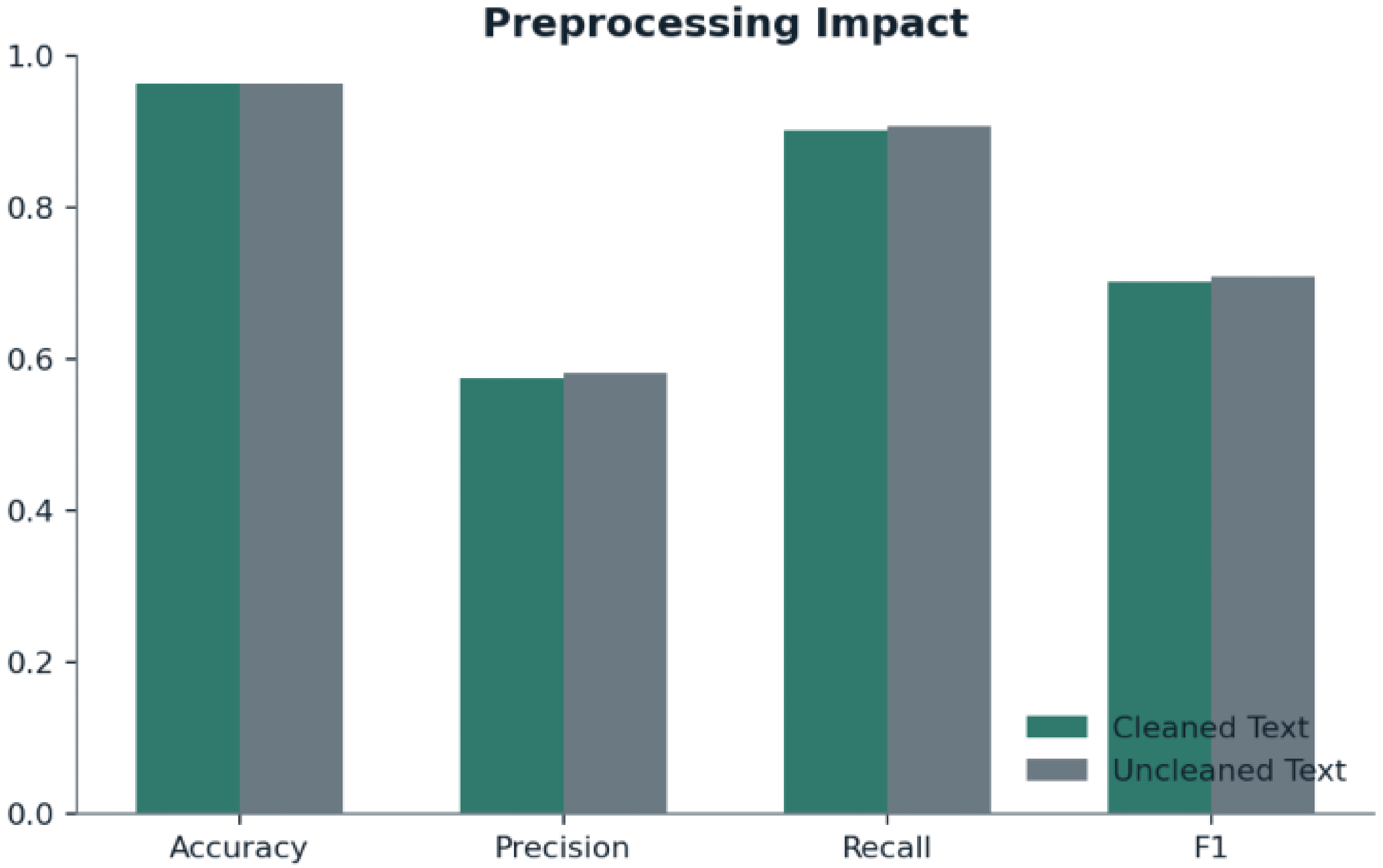

Figure 3: Structured Metadata Signals vs Fraud (Logo, Questions, Telecommuting)

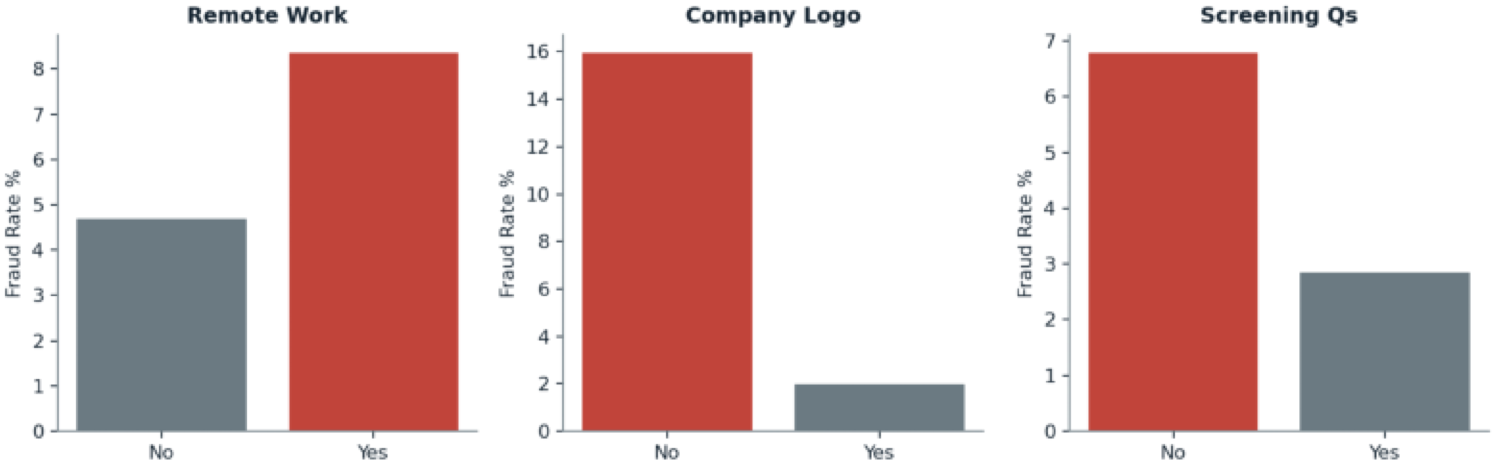

Figure 4: Job Description Length Distributions (Real vs Fraudulent)

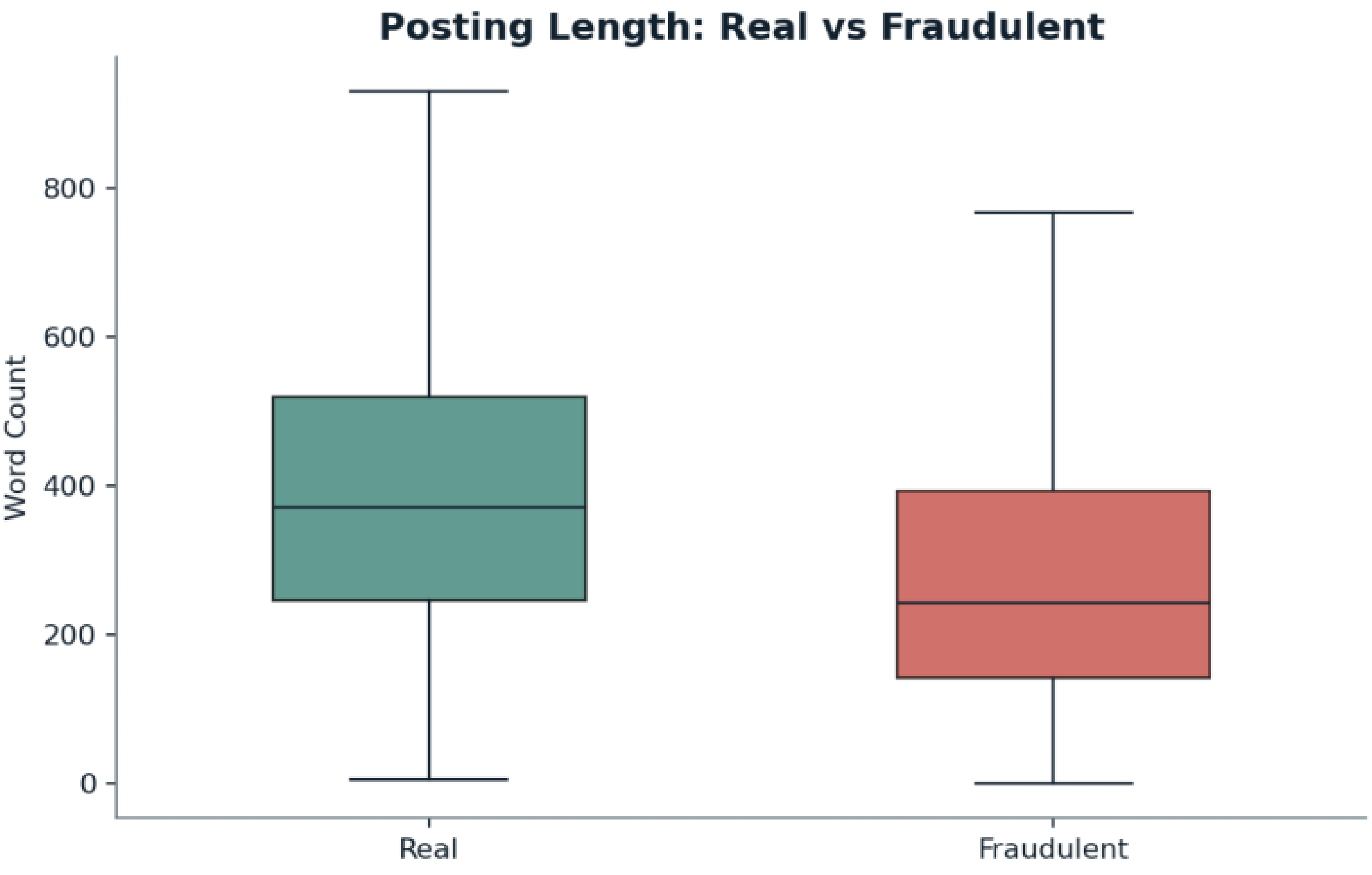

Figure 5: Model Performance Comparison (Accuracy, Precision, Recall, F1)

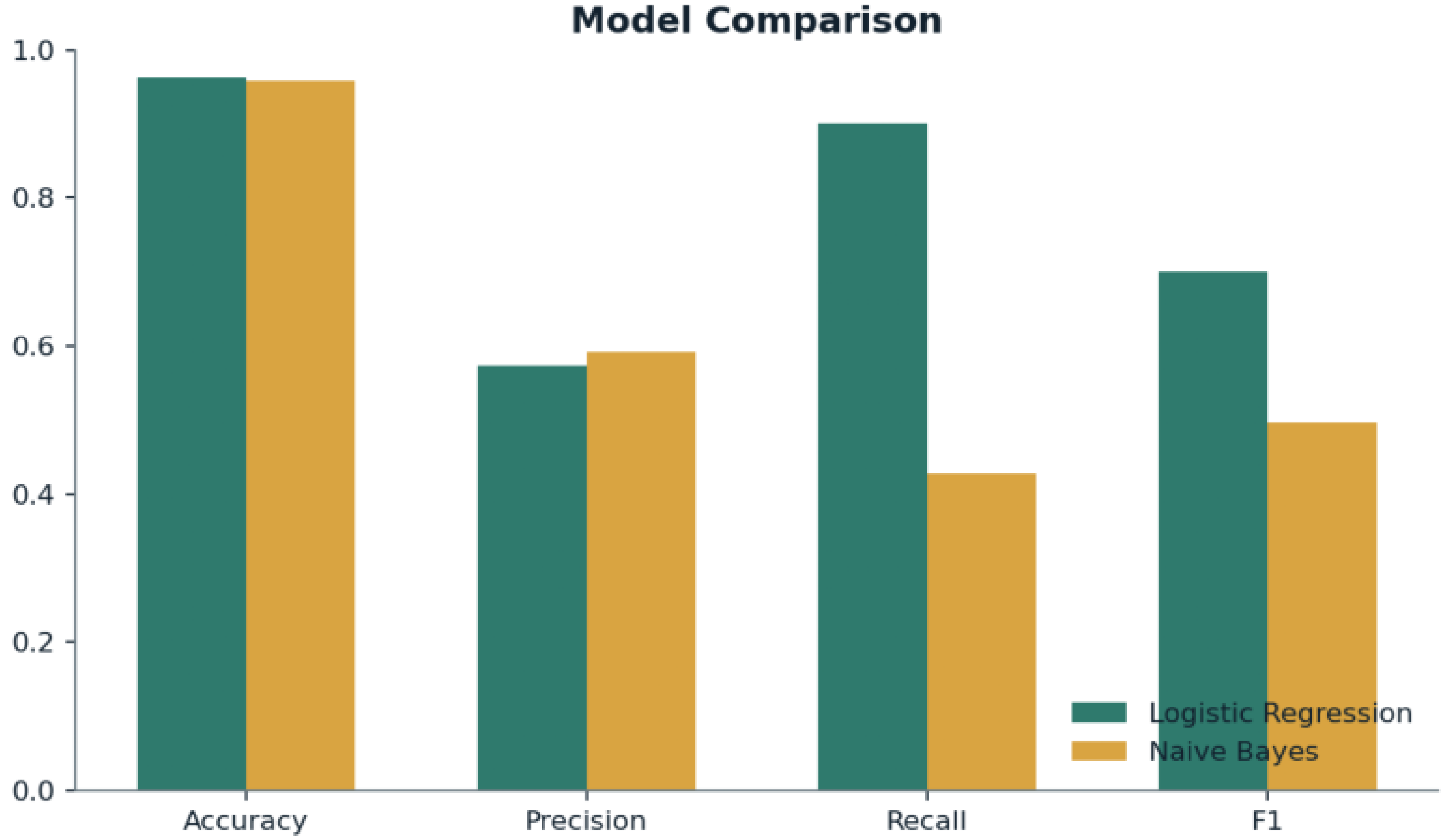

Figure 6: Confusion Matrix - Logistic Regression (Best Model)